AB CLOUD QUANTUM RESEARCH · MONOGRAPH SERIES

VOLUME Appendix

Appendix: Methods, Data, Reproducibility

Protocols of the series V1–V18: lattice, unfolding, statistics, the trace pipeline, certificates, thresholds and machine floors

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Contents

A.1 Lattice and gates **2**

A.2 Statistics and references **2**

A.3 Certificates and thresholds **2**

A.4 Reproducibility **3**

A.1 Lattice and gates

The AB-cloud Hamiltonian: a Hermitian Hofstadter-type L×L lattice at half flux ($\alpha = ½$ — the self-dual line of the discretized massless Dirac equation), vortex phases θ_k = atan2, on-site disorder W. Vortex protocol: cell centers (ix+0.5, iy+0.5), independent sampling, neutral charges. Series master seed 96; vortex RNG — seed + 3500, disorder — seed + 3501. The full handle space (V16): charge signs, vortex profile (site/midpoint), anisotropy t_x/t_y, staggered mass, twists (φx, φy), flux shift $\delta\alpha$ — a 96-configuration factorial design.

Spectral pipeline: eigvalsh (driver evr) → central band 60% (fld arithmetic) → polynomial unfolding (degrees 6→3, piecewise-linear fallback) → 200-level window. Bit-exact reproduction of the reference C3 protocol: 1.45e-16.

A.2 Statistics and references

The vector $S = (\langle r\rangle, \Sigma^2(L), \Delta_3(L), K(t), D_{KS})$ is pre-registered. The GUE reference for D_{KS} is empirical (an ensemble of 16 matrices 576×576, a pooled sample of 3168 $\{r_n\}$ values); reference ladders: Poisson $2\ln 2-1 \approx 0.386$, GOE ≈ 0.531, GUE ≈ 0.600. The ζ zeros — the Odlyzko file of 50,000 ordinates (SHA-256 in data/README.md); the smooth count $\bar{N}(\gamma) = \gamma/2\pi\cdot\ln(\gamma/2\pi e) + 5/8$. The algebraic function of the critical line (V16): $Z_{id}(t) = e^{i\theta(t)}\zeta(½+it)$, $\theta(t) = \mathrm{Im}\ln\Gamma(¼+it/2) - t\cdot\ln\sqrt{\pi}$; the Γ-closures $|\Gamma(½+iu)|^2 = \pi/\cosh\pi u$, $|\Gamma(iu)|^2 = \pi/(u\cdot\sinh\pi u)$, the Legendre product $|\Gamma(¼+iu)|\cdot|\Gamma(¾+iu)|$ — residuals ~1.3e-39 at dps 40.

The trace pipeline: Q(a) — a spline of the mpmath grid [0, 1.5] (601 nodes, the pole a = ½ shifted); the prime part — the Λ(n) sieve to $2\cdot10^6$ (cached), tail ≤ 1e-6; the smooth part — exact Si/Ci forms; PC0 — the recipe $u\cdot\Sigma\, e^{-u\cdot\bar{N}(\gamma)} - (1-e^{-uX})$, $U \in [0.05, 1.0]$, 40 points.

A.3 Certificates and thresholds

The interval layer: Gershgorin in fractions (limit_denominator 1e12), residuals in mpmath dps = 40, the gap exact. The data license layer: the zero file is the only external data; all verdicts are committed and independent of fresh recomputations.

The threshold table (threshold → fact): hermiticity 1e-12 → 0.0; residuals 1e-11 → 8.8e-15; M0 1e-6 → 4.6e-8; M1 5e-5 → 6.9e-6; PC0 0.99 → 0.9996; Th4 1e-9 → 4.4e-11; reduction 1e-9 → 7.1e-15; unitarity 1e-12 → 2.2e-16; blind z > 3 → 0.09; V18 gates 1e-9 → 2e-16; V18 entanglement > 0.5 → 0.984; V17 energy 1e-12 → 1.8e-15.

A.4 Reproducibility

Self-build: make tour (a tour over the committed verdicts), make test (pytest), make ivp (independent verification), make figures / monographs / workbook (artifact regeneration), make all. A fresh full recomputation of V1–V18: python3 scripts/compute_all.py and python3 scripts/compute_v161718.py (~10 minutes total; the zero file is included in the package). The web lab — web_lab/index.html, a self-contained file without external dependencies, including a playable AB-Pong.

Every campaign is deterministic: fixed seeds, cached sieves, fixed grids. A divergence of a fresh run from the committed verdicts is a release stopper (the test_repro test).

The appendix records all protocol decisions needed for an independent reproduction of the series V1–V18.